National
Qualifications
2022

2022 Biology

National 5

Finalised marking instructions

Marking instructions for each question

Section 1

Question	Response	Mark
1.	A	**1**
2.	B	**1**
3.	C	**1**
4.	A	**1**
5.	D	**1**
6.	A	**1**
7.	D	**1**
8.	B	**1**
9.	C	**1**
10.	B	**1**
11.	A	**1**
12.	D	**1**
13.	C	**1**
14.	C	**1**
15.	B	**1**
16.	B	**1**
17.	B	**1**
18.	C	**1**
19.	A	**1**
20.	D	**1**
21.	D	**1**
22.	A	**1**
23.	C	**1**
24.	D	**1**
25.	C	**1**

Section 2

Question			Expected response	Max mark	Additional guidance
1	(a)	(i)	Cell wall	1	Not acceptable: wall on its own
		(ii)	50	1	
	(b)		They would not contain any chloroplasts	1	Not acceptable: contains less chloroplasts **OR** don't contain chlorophyll
2	(a)		Double (stranded) helix	1	Not acceptable: helix on its own
	(b)		Y	1	
	(c)	(i)	Protein	1	Not acceptable: amino acids
		(ii)	Different (order/sequence of) amino acid(s) **OR** Different shape/structure (of protein/molecule) **OR** The function (of the protein/ molecule) could be different/ affected	1	Not acceptable: it would be different; different protein
		(iii)	Radiation/(some) chemicals **OR** suitable examples	1	Not acceptable: radioactivity
3.	(a)		1.7	1	
	(b)	(i)	Enzyme/pepsin is less active at pH7/(the) high pH **OR** Enzyme/pepsin is more active at pH 2/(the) low pH **OR** Enzyme/pepsin is not at its optimum pH in test tube B/pH 7 **OR** Enzyme/pepsin is at its optimum pH in test tube A/pH 2	1	Not acceptable: enzyme doesn't work as well

Question			Expected response	Max mark	Additional guidance
3.	(b)	(ii)	Not the specific/complementary enzyme/active site **OR** Catalase/the enzyme doesn't break down egg white/protein **OR** Catalase/the enzyme has no effect on egg white **OR** Egg white is not the substrate for catalase/the enzyme	1	Not acceptable: wrong/not correct enzyme
	(c)	(i)	Shape would change/be different	1	Not acceptable: denatured on its own without a description structure/it would change
		(ii)	Decreases/stops/slows down	1	
4.			**Similarities** 1. glucose is converted to pyruvate. **(1)** 2. completed/take place in the cytoplasm. **(1)** 3. yields two (molecules of) ATP. **(1)** 4. (pathways) occur without/in the absence of oxygen. **(1)** 5. is enzyme controlled. **(1)** **Max 2 marks** **Differences** 6. The end products in yeast/fungal cells are ethanol and CO_2 **(1)** 7. In muscle/animal cells the product is lactate. **(1)** **OR** 8. In one, the end products are ethanol and CO_2 and the other is lactate. **(1)** **Max 2 marks**	4	Arrows as alternatives are acceptable in points 1, 3, 6, 7, 8. Arrows must have arrow heads → 'Plant cells' on its own is not acceptable but 'plants and yeast' does not negate. If plant cells (but not yeast) are referred to throughout, then this should only be penalised once. Point 8 can only be awarded if neither points 6 or 7 are awarded.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>5.</td><td>(a)</td><td></td><td><table><tr><th>Structure</th><th>Letter</th></tr><tr><td>Bacterial cell</td><td>E</td></tr><tr><td>Gene for penicillin</td><td>C</td></tr><tr><td>(modified) plasmid</td><td>D</td></tr></table>
1 mark for each correct answer</td><td>3</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>Insert (modified) plasmid/D into a (different) bacterial/host cell</td><td>1</td><td>If an incorrect structure for D in 5a is carried forward then this would not negate an otherwise correct response for 5b.
Not acceptable:
Any implication that the plasmid is returned to original bacterial cell.</td></tr>
<tr><td>6.</td><td>(a)</td><td></td><td>So that only one variable/factor was being altered
OR
To make the experiment valid/ ensure validity
OR
To allow a valid comparison</td><td>1</td><td>Not acceptable: to make it fair/ a fair test; to make results valid
Any additional incorrect answer eg reliable, would negate</td></tr>
<tr><td></td><td>(b)</td><td></td><td>60</td><td>1</td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td>Live/living peas carry out aerobic respiration/respire
OR
Dead peas do not carry out aerobic respiration/respire</td><td>1</td><td>Answer must refer to peas</td></tr>
<tr><td></td><td>(d)</td><td></td><td>Repeat it/investigation/experiment</td><td>1</td><td></td></tr>
<tr><td>7.</td><td>(a)</td><td>(i)</td><td>Spindle (fibres)</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Nuclear membrane(s) form/develop
OR
(Two) nuclei form</td><td>1</td><td>Any additional subsequent stage, provided the order is correct, would not negate.
Not acceptable:
any reference to chromatids
nuclear membrane ‘reforms’</td></tr>
<tr><td></td><td>(b)</td><td></td><td>Unspecialised</td><td>1</td><td>Also acceptable:
non-specialised/undifferentiated</td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
8.	(a)		He is heterozygous/Aa/has both alleles/both forms of the gene **AND** has FH/ familial hypercholesterolemia/the condition/is affected	**1**	Not acceptable: he has high cholesterol
	(b)		100	**1**	
	(c)		discrete	**1**	
9.	(a)		ovary	**1**	
	(b)		diploid	**1**	
	(c)	(i)	5	**1**	
		(ii)	35	**1**	
	(d)		Mitochondrion/mitochondria	**1**	
10.	(a)	(i)	Hormones	**1**	Not acceptable: Named examples alone
		(ii)	Target tissues/cells have <u>receptors</u>/<u>receptor</u> proteins **(1)** which are complementary/specific to a hormone/chemical messenger **OR** Hormones are specific to the target tissue/cells **(1)**	**2**	Must be clear that the receptors are on the target tissues.
	(b)		Phloem/xylem	**1**	
11.	(a)		Gas - Carbon dioxide **(1)** Method of transport – Diffusion **(1)**	**2**	Not acceptable: Passive transport (but would not negate)
	(b)		Thin walls/walls only one cell thick/ large surface area	**1**	Not acceptable: One cell thick (alone); Extensive blood supply

Question			Expected response	Max mark	Additional guidance
12.	(a)		Disease-causing micro-organism/ microbe/virus/bacteria/fungi	**1**	
	(b)	(i)	As the vaccine uptake decreases, the number of cases increases (or converse) **OR** As fewer people get vaccinated, more people get measles (or converse) **OR** As the number of people getting the vaccine decreases, the number of cases of measles increases (or converse)	**1**	Also acceptable: Any other suitable equivalent
		(ii)	People believed that the threat/risk (of catching measles) was less **OR** the UK was measles-free/it was less common **OR** (People reluctant to have children immunised as) the MMR vaccine was associated with autism	**1**	Also acceptable: there are more 'anti-vaxxers'
	(c)		2:7	**1**	
	(d)		92% vaccinated but needs to be (at least) 95% to be measles-free. **OR** Less than 95% vaccinated	**1**	Values required. Not acceptable: only 92% were vaccinated.
13.	(a)		(Bubble) potometer	**1**	
	(b)		As total leaf surface area decreases the rate of water uptake decreases **OR** As total leaf surface area increases the rate of water uptake increases	**1**	
	(c)	(i)	Transpiration	**1**	
		(ii)	Guard (cells)	**1**	

Question			Expected response	Max mark	Additional guidance
14.	(a)	(i)	Volume of water/temperature/pH/ time left for/age of seed/moisture level	**1**	Not acceptable: spacing of seeds/size of dish/light/ CO_2 concentration
		(ii)	90	**1**	
		(iii)	It has the lowest percentage of germinating seeds	**1**	Not acceptable: it has the lowest number/amount of germinating seeds/fewer seeds germinated in D
	(b)		Intraspecific	**1**	
	(c)		Resources are limited/in short supply	**1**	Acceptable: named resource(s)
15.	(a)		Light meter **(1)** Don't cast a shadow/don't block the light/ensure sensor is facing light source **(1)**	**2**	Not acceptable: light intensity meter; light/moisture meter Award one mark if wrong apparatus given but correct precaution for that apparatus is given.
	(b)	(i)	May have been duller/cloudier (on day 2) **OR** less sunlight/sun/light (on day 2) **OR** Days 1 and 3 may have been brighter/less cloudy/sunnier (must be comparative)	**1**	Not acceptable: Different weather/light may have been blocked/no sun
		(ii)	To reduce the effect of atypical results **OR** Improve the reliability (of the results)	**1**	Any reference to accuracy/validity would negate Not acceptable: Reliability/improve reliability of experiment/so an average can be calculated (but this would not negate)
	(c)		(soil) moisture/pH/temperature/ humidity/wind speed	**1**	
16.	(a)		Oxygen	**1**	
	(b)		Light to chemical (energy) **OR** Light (energy) $\rightarrow$ Chemical (energy)	**1**	
	(c)		M – <u>light intensity</u> **(1)** N – temperature **(1)**	**2**	Not acceptable: carbon dioxide concentration

Question			Expected response	Max mark	Additional guidance
17.	(a)		Scale and label on y-axis **(1)** All bars at correct height and are able to be identified by shading/labels **(1)**	**2**	
	(b)		Invalid Reason: it is not true for *V. cholerae*/all species **OR** Ethanol was more effective at preventing growth for only *S. flexneri* and *S.paratyphi* **OR** Methanol was more effective at preventing growth for *V. cholerae* **OR** Growth of V. cholerae was greater with ethanol than methanol	**1**	
	(c)		Bacterial growth would increase/more bacteria grow **OR** Growth of bacteria/it would not be reduced as much	**1**	Also acceptable: twice as many bacteria would grow Not acceptable: faster growth of bacteria
18.	(a)	(i)	To make proteins/amino acids	**1**	
		(ii)	0.6	**1**	
	(b)	(i)	(Algal) bloom	**1**	
		(ii)	Blocks light/sunlight/reduces light levels **(1)** Plants can't photosynthesise/carry out (less) photosynthesis **(1)**	**2**	Not acceptable: 'Blocks the sun' on its own Cause of plant death must be clearly linked to only lack of photosynthesis.

[END OF MARKING INSTRUCTIONS]

General marking principles for National 5 Biology

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the marking instructions for each question, which identify the key features required in candidate responses.

(a) Marks for each candidate response must **always** be assigned in line with these general marking principles and the detailed marking instructions for this assessment.

(b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error at an early stage in the first part of a question, credit should normally be given for subsequent answers that are correct with regard to this original error. Candidates should not be penalised more than once for the same error.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units, if appropriate) on its own.

(g) In the detailed marking instructions, if a word is <u>underlined</u> then it is essential; if a word is (bracketed) then it is not essential.

(h) In the detailed marking instructions, words separated by / are alternatives.

(i) A correct answer can be negated if:
- an extra, incorrect, response is given
- additional information that contradicts the correct response is included.

(j) Unless otherwise required by the question, use of abbreviations (eg DNA, ATP) or chemical formulae (eg CO_2, H_20) are acceptable alternatives to naming.

(k) Where incorrect spelling is given:
- If the correct word is recognisable then give the mark.
- If the word can easily be confused with another biological term then do not give the mark eg mitosis and meiosis.
- If the word is a mixture of other biological words then do not give the mark, eg osmotis, respirduction, protosynthesis.

(l) Presentation of data
- If a candidate provides two graphs or charts, mark both and give the higher score.
- If a question asks for a particular type of graph and the wrong type is given, then full marks cannot be awarded. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label.
- If the x and y data are transposed, then do not give the scale and label mark.
- If the graph uses less than 50% of the axes, then do not give the scale and label mark.
- If 0 is plotted when no data is given, then do not give the plot mark (ie candidates should only plot the data given).
- No distinction is made between bar graphs and histograms for marking purposes.
- In a pie chart lines must originate from the central point and extend to tick marks. Labels must be given in full.

(m) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only answer or present in brief form;
- **describe**, they must provide a statement as opposed to simply one word;
- **explain**, they must provide a reason for the information given;
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined;
- **calculate**, they must determine a number from given facts, figures or information;
- **predict**, they must indicate what may happen based on available information;
- **suggest**, they must apply their knowledge and understanding to a new situation.

2022 National 5 Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**A**	**56**	16	6	21	1	0
2	**B**	20	**60**	12	7	1	0
3	**C**	12	12	**66**	10	1	0
4	**A**	**71**	14	6	8	1	0
5	**D**	3	9	39	**48**	1	0
6	**A**	**40**	35	9	16	1	0
7	**D**	11	15	6	**67**	1	0
8	**B**	11	**83**	3	3	1	0
9	**C**	4	13	**64**	18	1	0
10	**B**	13	**51**	29	6	1	0
11	**A**	**64**	17	13	5	1	0
12	**D**	3	19	12	**65**	1	0
13	**C**	20	20	**41**	19	1	0
14	**C**	13	18	**46**	23	1	0
15	**B**	19	**56**	6	17	1	0
16	**B**	31	**40**	16	12	1	0
17	**B**	6	**80**	7	7	1	0
18	**C**	2	6	**87**	4	1	0
19	**A**	**94**	1	3	1	1	0
20	**D**	1	3	30	**65**	1	0
21	**D**	13	28	7	**51**	1	0
22	**A**	**40**	39	8	13	1	0
23	**C**	13	18	**57**	11	1	0
24	**D**	2	5	17	**76**	1	0
25	**C**	6	43	**33**	17	1	0

2022 Nat5 Biology Section 2 Marks Data Analysis

2022 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.88	88%	20	1%
Q1aii	1	0.50	50%	45	4%
Q1b	1	0.42	42%	51	13%
Q2a	1	0.92	92%	30	1%
Q2b	1	0.88	88%	29	0%
Q2ci	1	0.30	30%	38	5%
Q2cii	1	0.14	14%	38	13%
Q2ciii	1	0.44	44%	54	5%
Q3a	1	0.37	37%	30	8%
Q3bi	1	0.12	12%	43	5%
Q3bii	1	0.26	26%	45	7%
Q3ci	1	0.19	19%	34	9%
Q3cii	1	0.47	47%	53	10%
Q4	4	2.27	57%	65	12%
Q5a	3	2.15	72%	58	1%
Q5b	1	0.27	27%	46	13%
Q6a	1	0.08	8%	26	1%
Q6b	1	0.42	42%	55	5%
Q6c	1	0.38	38%	49	5%
Q6d	1	0.50	50%	26	4%
Q7ai	1	0.78	78%	51	8%
Q7aii	1	0.11	11%	28	8%
Q7b	1	0.51	51%	57	13%
Q8a	1	0.16	16%	40	8%
Q8b	1	0.42	42%	57	5%
Q8c	1	0.54	54%	55	12%
Q9a	1	0.85	85%	32	2%
Q9b	1	0.85	85%	37	0%
Q9ci	1	0.64	64%	47	3%
Q9cii	1	0.54	54%	52	5%
Q9d	1	0.44	44%	59	12%
Q10ai	1	0.57	57%	59	5%
Q10aii	2	0.56	28%	63	14%
Q10b	1	0.50	50%	53	9%
Q11a	2	0.82	41%	50	4%
Q11b	1	0.52	52%	52	13%
Q12a	1	0.08	8%	29	14%
Q12bi	1	0.61	61%	47	6%
Q12bii	1	0.78	78%	32	3%
Q12c	1	0.20	20%	32	6%
Q12d	1	0.19	19%	33	4%
Q13a	1	0.15	15%	38	29%
Q13b	1	0.15	15%	47	6%
Q13ci	1	0.34	34%	58	6%
Q13cii	1	0.69	69%	58	8%
Q14ai	1	0.64	64%	34	3%
Q14aii	1	0.71	71%	49	7%
Q14aiii	1	0.20	20%	35	5%
Q14b	1	0.69	69%	55	8%
Q14c	1	0.31	31%	37	4%
Q15a	2	0.99	50%	45	7%
Q15bi	1	0.43	43%	20	4%
Q15bii	1	0.29	29%	30	3%
Q15c	1	0.89	89%	29	2%
Q16a	1	0.71	71%	43	2%
Q16b	1	0.27	27%	53	15%
Q16c	2	0.98	49%	49	7%
Q17a	2	1.30	65%	41	3%
Q17b	1	0.06	6%	25	2%
Q17c	1	0.51	51%	41	6%
Q18ai	1	0.21	21%	51	8%
Q18aii	1	0.57	57%	53	19%
Q18bi	1	0.70	70%	44	13%
Q18bii	2	0.73	37%	46	5%
Paper	75	35.14	47%		

2022 Nat5 Biology Section 2 Distribution of Marks Analysis

2022 Item	0	1	2	3	4
Q1ai	12	88			
Q1aii	50	50			
Q1b	58	42			
Q2a	8	92			
Q2b	12	88			
Q2ci	70	30			
Q2cii	86	14			
Q2ciii	56	44			
Q3a	63	37			
Q3bi	88	12			
Q3bii	74	26			
Q3ci	81	19			
Q3cii	53	47			
Q4	26	10	13	14	38
Q5a	7	15	33	45	
Q5b	73	27			
Q6a	92	8			
Q6b	58	42			
Q6c	62	38			
Q6d	50	50			
Q7ai	22	78			
Q7aii	89	11			
Q7b	49	51			
Q8a	84	16			
Q8b	58	42			
Q8c	46	54			
Q9a	15	85			
Q9b	15	85			
Q9ci	36	64			
Q9cii	46	54			
Q9d	56	44			
Q10ai	43	57			
Q10aii	62	21	18		
Q10b	50	50			
Q11a	40	38	22		
Q11b	48	52			
Q12a	92	8			
Q12bi	39	61			
Q12bii	22	78			
Q12c	80	20			
Q12d	81	19			
Q13a	85	15			
Q13b	85	15			
Q13ci	66	34			
Q13cii	31	69			
Q14ai	36	64			
Q14aii	29	71			
Q14aiii	80	20			
Q14b	31	69			
Q14c	69	31			
Q15a	30	41	29		
Q15bi	57	43			
Q15bii	71	29			
Q15c	11	89			
Q16a	29	71			
Q16b	73	27			
Q16c	39	25	37		
Q17a	17	36	47		
Q17b	94	6			
Q17c	49	51			
Q18ai	79	21			
Q18aii	43	57			
Q18bi	30	70			
Q18bii	48	32	21		